Jinglan (Bella) DAI
(+1) 314-814-7426 | bella.d@wustl.edu | belladai36.github.io | 6515 Wydown Blvd. | 63105 St. Louis, MO

EDUCATION

Washington University in St. Louis, St. Louis, MO 08/2024 - 07/2028
Bachelor of Arts, Major: Statistics and Economics, Minor: Classics GPA: 3.99/4.00

- Coursework: Machine Learning, Bayesian Statistics, Linear Algebra, Probability, Linear Regression, Econometrics, Principles of Microeconomics, Principles of Macroeconomics, Latin
- Fall 2024, Spring 2025, Fall 2025, Spring 2026 – Dean's List in the College of Arts & Sciences

RESEARCH INTERESTS

Conformal Prediction and Uncertainty Quantification; Survival Analysis; High-dimensional Statistical Inference; Statistical Learning; Biostatistics; Reproducible Statistical Computing

RESEARCH EXPERIENCE

Undergraduate Independent Study: *Two-Layer Hierarchical Conformal Prediction*, St. Louis, MO 01/2026 – Present

- Studied hierarchical conformal prediction and survival-analysis methods for dependent, right-censored time-to-event outcomes;
- Developed calibration procedures and finite-sample coverage guarantees for cluster- and individual-level lower prediction bounds;
- Extended the framework toward conditional and group-conditional coverage under within-cluster dependence, censoring, covariate shift, and unequal cluster sizes.

Research Project*: Statistical Learning for a Wolffia Single-Cell Atlas*, St. Louis, MO 06/2026 – Present

- Developed a reproducible PIP-seq/Scanpy workflow for quality control, normalization, feature selection, dimensionality reduction, Leiden clustering, UMAP visualization, and trajectory inference;
- Built a cross-dataset statistical-learning framework using feature harmonization, latent program scores, supervised classification, and quantitative evaluation of label-transfer performance.

Statistical Learning: *High-Dimensional Classification*, St. Louis, MO 11/2025 – 01/2026

- Analyzed high-dimensional predictor data from 72 patients for binary subtype classification;
- Compared logistic regression, naive Bayes, discriminant analysis, decision trees, random forests, boosting, K-nearest neighbors, and support vector machines;
- Applied feature screening and out-of-sample model validation to assess overfitting, predictive accuracy, and generalization performance.

Research Assistant, WashU Özpolat's Lab: *Statistical Analysis*, St. Louis, MO 06/2026 – Present

- Reviewed single-cell methods to define candidate labels, marker features, and reproducible annotation criteria;
- Conducted exploratory data analysis in CELLxGENE and Jupyter, evaluating expression distributions, candidate features, and annotation consistency with domain experts.

Research Assistant, WashU Cox Lab: *Quantitative Image Analysis,* St. Louis, MO 01/2026 – Present

- Standardized experimental protocols and measurement procedures to support reproducible data collection;
- Built quantitative image-analysis workflows that transformed high-resolution images into structured features for statistical comparison and longitudinal trend analysis;
- Maintained data quality through protocol standardization, measurement validation, reproducible records, and laboratory QA/QC.

Social Research on MBTI Heat, *Initiator & Team Leader*, Wuhan, China 11/2022 – 05/2023
China Thinks Big Project, Qualified for 2023 Global Final

- Led a five-person research team investigating adolescents' reliance on MBTI personality labels;
- Collected and analyzed 118 survey responses using descriptive statistics to assess test objectivity and standardization;
- Integrated survey evidence with literature on psychological, social, and market drivers of user behavior;
- Integrated survey and literature findings into a research paper and qualified for the 2023 China Thinks Big Global Final.
- Created a five-minute interactive data-storytelling video communicating findings to a nontechnical audience.

Statistical Modelling: *CO_2 Emission and Global Temperature Change*, *Member* 11/2022 – 02/2023

- Fit polynomial-regression, indicator-variable, and ARIMA models in Python to analyze long-term trends and generate forecasts through 2050 and 2100.
- Performed correlation and regression analysis to quantify the relationship between atmospheric CO2 and global temperature change.

Jinglan (Bella) DAI
(+1) 314-814-7426 | bella.d@wustl.edu | belladai36.github.io | 6515 Wydown Blvd. | 63105 St. Louis, MO

PROFESSIONAL EXPERIENCE

Data Analyst Intern, *Gala Circle Inc.*, Remote 06/2026 – 09/2026

- Collected, cleaned, and validated restaurant revenue, POS fees, labor costs, outreach, and conversion data;
- Segmented merchants by revenue and operational needs, combined quantitative findings with interview data, and delivered targeted operational and promotional recommendations.

WashU Residential Advisor, St. Louis, MO 08/2026 – Present

- Supported residents' academic, social, and personal well-being while maintaining an inclusive and safe residential community;
- Managed on-call incidents, documented concerns, and coordinated timely responses with professional staff;
- Planned community programs and connected residents with relevant campus resources.

WashU Department of Statistics and Data Science, St. Louis, MO 08/2025 – 12/2025

- *Graded coursework for SDS 4010: Probability with consistent rubric-based evaluation.*
- *Tutored students in probability and statistics, translating technical concepts into clear problem-solving steps.*

WashU Arts & Sciences Pre-College Program, *Program Assistant,* St. Louis, MO 04/2025 – 07/2025

- Coordinated academic and college-readiness activities, communicated program policies, and supported students in a fast-paced residential program.

WashU Office for International Student Engagement, *WU GUIDE,* St. Louis, MO 04/2025 – 08/2025

- Mentored new international students through campus orientation and academic and social transition.

EXTRACURRICULAR EXPERIENCE

Adopt A Grandparent (AAG), *Treasurer*, St. Louis, MO 08/2024 - Present

- Volunteer weekly at two local retirement communities;
- Manage budgeting, funding, and expense records while supporting organizational financial planning.

SKILLS AND INTERESTS

- **Languages**: Chinese Mandarin (Native), English (Proficient), Latin (Familiar), Korean (Beginner)
- **Technical Skills:** Python (Proficient), R (Proficient), SQL (Intermediate), Excel (Intermediate), JavaScript (Basic), LaTeX (Basic); pandas, NumPy, scikit-learn, Scanpy, AnnData, Jupyter, Git/GitHub.